Multi-Disease Detection Report

CNN + LLM contextual reasoning • generated 2026-08-24 22:55

Patient context

Age	54
Sex	Female
Symptoms	Abnormal vaginal bleeding after intercourse, bleeding between periods, pelvic pain, unusual vaginal discharge
History	45-year-old woman, persistent high-risk HPV infection, irregular cervical screening history

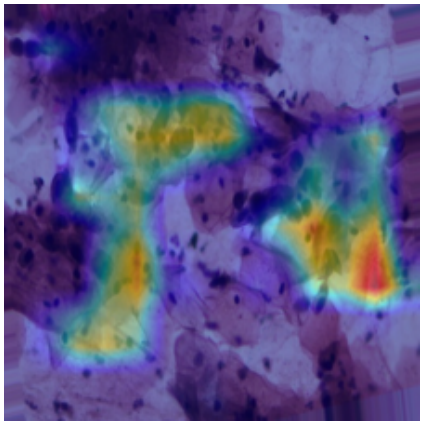
Model prediction

cervix_koilocytotic **Precursor**

Cervical Cytology (SIPaKMeD) · Cervix / Gynecologic · Pap smear cytology

#	Condition	Confidence
1	cervix_koilocytotic	90.4%
2	kidney_tumor	2.2%
3	lung_adenocarcinoma	1.0%
4	brain_meningioma	0.8%
5	lung_squamous_cell_carcinoma	0.8%

Grad-CAM (model attention)



LLM contextual interpretation

Model suggests Cervical Cytology (SIPaKMeD) (cervix_koilocytotic) at 90.4% confidence.

Model finding. The CNN's top class is 'cervix_koilocytotic' (Precursor, Cervix / Gynecologic) on Pap smear cytology, at 90.4%.

Confidence. High model confidence (90.4%) with a clear 88.2 pt margin over the runner-up.

Differential

- **cervix_koilocytotic** — ranked #1 at 90.4%
- **kidney_tumor** — ranked #2 at 2.2%
- **lung_adenocarcinoma** — ranked #3 at 1.0%

Grad-CAM correlation. Model attention was a moderately sized region in the central portion of the image (~11% high-salience area). Verify this region is anatomically plausible for Pap smear cytology.

Biomedical context. [KB] Koilocytes — perinuclear halos indicating HPV cytopathic effect. Group: Cervical Cytology (SIPaKMeD); organ: Cervix / Gynecologic; modality: Pap smear cytology; category: Precursor (severity: moderate). Cervical Pap cytology classifies squamous cells from normal (superficial/intermediate, parabasal) to abnormal (koilocytotic, dyskeratotic), aiding HPV-related lesion screening. [BioGPT fallback — set ENABLE_BIOGPT=true to run the real model] Abnormal cytology is followed by HPV testing and, if indicated, colposcopy with directed biopsy per screening guidelines. [Med-PaLM 2: PLACEHOLDER — no public API; see medpalm_adapter.py to wire MedLM]

Patient considerations. Provided context (age: 54; sex: Female; symptoms: Abnormal vaginal bleeding after intercourse, bleeding between periods, pelvic pain, unusual vaginal discharge; history: 45-year-old woman, persistent high-risk HPV infection, irregular cervical screening history) should be correlated with the finding.

Recommended next steps

- Correlate with the full clinical picture and prior imaging.
- Obtain gold-standard confirmation appropriate to the modality (e.g., biopsy/histology or specialist review).
- Escalate to the relevant specialist if the category is malignant or high-severity.

Uncertainty & limitations

- Interpretation generated WITHOUT the Claude reasoning model (offline fallback).
- Model trained on specific public datasets; performance may not transfer to other scanners/populations.
- A confident softmax score is not calibrated probability of disease.

■ Research decision-support only — not a medical diagnosis. All findings require confirmation by a licensed clinician.